A Structural Selection Principle for Dynamically Consistent Field Configurations

Maximum-Entropy Weighting on Physical Solution Spaces

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Abstract

Not every mathematically valid solution of fundamental field equations is necessarily realised in nature. We propose a two-level framework in which the fundamental dynamics defines the space of *possible* field configurations via the stationarity condition $\delta S[\Phi] = 0$, while a secondary structural selection principle identifies the *physically preferred* configurations within that solution space. The selection functional $\mathcal{E}[\Phi]$ is constructed from five covariant structural diagnostics measuring residual stability, conservation integrity, dynamical activity, inter-field connectivity, and theoretical consistency. We derive the weights of the selection functional from a Maximum-Entropy principle on the solution manifold, yielding a Boltzmann-type distribution over field configurations. A toy-model numerical test on a one-dimensional scalar lattice illustrates that the selection principle can favour structured, topologically stable configurations over generic solutions in a reduced setting. In addition, a reproducible FLRW benchmark compares three scalar–tensor backgrounds — a static de Sitter solution, a dynamically active slow-roll configuration, and a ghost-near stress test — demonstrating that the functional ranks these classes in physically expected order. The framework is presented as a research programme rather than a complete theory; its physical interpretation and open mathematical questions are discussed.

1 Introduction

A long-standing question in theoretical physics concerns the status of solutions to fundamental field equations: given a rich space of mathematically valid solutions, what determines which solutions describe physical reality?

Standard approaches address this question either through initial conditions, anthropic arguments, or symmetry-based selection criteria. Here we explore a complementary possibility: that physical realisability is governed by a structural stability criterion that is independent of, but logically subsequent to, the dynamical equations themselves.

The central hypothesis of this paper is the following.

> **Structural Selection Hypothesis:** Among all solutions of the fundamental dynamics, those that are physically realised are *hypothesised* to be precisely those that minimise a covariant structural energy functional $\mathcal{E}[\Phi]$ measuring residual instability, conservation violation, dynamical inactivity, inter-field disconnection, and theoretical inconsistency.

This hypothesis does not replace the dynamical equations; it operates as a selection layer above them. The primary conceptual contribution of this work is not a new theory of gravity or matter, but the formulation of a covariant, mathematically grounded selection principle that can be applied to arbitrary field theories.

The paper is organised as follows. Section 2 introduces the two-level architecture. Section 3 defines the five structural diagnostics. Section 4 derives the selection functional from Maximum-Entropy principles. Section 5 presents a numerical proof-of-concept. Section 6 develops the FLRW background reduction. Section 7 discusses implications, limitations, and open questions. Section 8 concludes.

2 Two-Level Architecture

2.1 Level 1: Fundamental Dynamics

Let Φ denote a collection of fields (metric, matter fields, scalars, gauge fields). A configuration is *dynamically admissible* if it satisfies the Euler–Lagrange equations derived from the action $S[\Phi]$:

$$\delta S[\Phi] = 0. \tag{1}$$

This defines the *solution manifold* $\mathcal{M}_S = \{\Phi \mid \delta S[\Phi] = 0\}$, the space of all possible field configurations. We make no assumption about the specific form of S; the framework is intended to be applicable to any diffeomorphism-covariant field theory.

2.2 Level 2: Structural Selection

Among all $\Phi \in \mathcal{M}_S$, we propose that physically realised configurations are sampled according to a structurally weighted probability measure on $\mathcal{M}_S$. The fundamental object is therefore a distribution $P[\Phi]$ over the solution manifold, derived in Section 4 from a Maximum-Entropy principle.

In the limit of sharp structural constraints (large positive Lagrange multipliers λ_i), this distribution concentrates around its dominant support, and the probabilistic statement reduces to a variational one:

$$\Phi_{\text{phys}} = \underset{\Phi \in \mathcal{M}_S}{\arg\min}\ \mathcal{E}[\Phi] \qquad \text{(saddle-point approximation)}. \tag{2}$$

Equation (2) is therefore not an independent postulate but a derived consequence of the probabilistic framework at strong coupling. At finite λ_i, the framework is fundamentally probabilistic rather than purely variational.

We emphasise that this is a *selection principle*, not a modification of the dynamics. No new degrees of freedom are introduced. The diagnostics are evaluated *on* solutions of $\delta S = 0$, not incorporated into the action.

3 Five Structural Diagnostics

We define five covariant functionals on the configuration space of the theory, each vanishing (or attaining its optimal value) on exact, stable, well-connected solutions. The five diagnostics introduced below are not claimed to be unique. They should be understood as a *minimal covariant basis* of structural monitors capturing five conceptually distinct aspects of physical viability: equation-of-motion proximity, conservation integrity, regulated dynamical activity, coupling structure, and theoretical consistency. Table 1 provides a compact overview.

3.1 Residual Stability Functional

A physically realisable configuration must satisfy its own equations of motion. We measure deviations from the solution manifold by the covariant residual norm:

$$\mathcal{H}[\Phi] = -\sum_i \left(\frac{\delta S}{\delta \Phi_i}\right)^2 = -\|R[\Phi]\|^2, \qquad R_i \equiv \frac{\delta S}{\delta \Phi_i}. \tag{3}$$

By construction $\mathcal{H} \leq 0$, with $\mathcal{H} = 0$ if and only if all field equations are satisfied. $\mathcal{H}$ is closely related to Lyapunov-type stability functionals: defining $V[\Phi] = \|R[\Phi]\|^2 = -\mathcal{H}$, exact solutions correspond to the zero set of V, and V decreases monotonically under the auxiliary gradient flow

$$\partial_\tau \Phi_i = -\frac{\delta V}{\delta \Phi_i} = \frac{\delta \mathcal{H}}{\delta \Phi_i}, \tag{4}$$

making $\mathcal{H}$ a natural Lyapunov functional on configuration space. The gradient flow (4) is an auxiliary mathematical construction; V is not the Hamiltonian energy of the system.

3.2 Conservation Functional

Stable physical reality requires conservation laws. We measure local energy–momentum conservation violations by:

$$\mathcal{B}[\Phi] = -(\nabla_\mu T^{\mu\nu})(\nabla_\rho T^\rho{}_\nu), \tag{5}$$

a covariant squared-divergence monitor. $\mathcal{B} = 0$ when energy–momentum is exactly conserved; $\mathcal{B} < 0$ signals flux inconsistency. In any theory for which $\delta S/\delta g^{\mu\nu} = 0$ holds exactly, $\nabla_\mu T^{\mu\nu} = 0$ follows from the contracted Bianchi identity and $\mathcal{B} = 0$ automatically. $\mathcal{B}$ is therefore most informative — and non-trivially diagnostic — for approximate, truncated, coarse-grained, or numerically reconstructed solutions, where such identities may not hold exactly.

3.3 Dynamical Activity Functional

A physically interesting universe must sustain non-equilibrium structure production. This is motivated by the observation, following Prigogine [5], that stable complex structures require regulated dissipative activity rather than thermodynamic equilibrium: a completely static universe cannot generate the structured configurations we observe. For a scalar field C we define a bounded, dimensionless activity measure:

$$\mathcal{S}_C[\Phi] = \eta_C \frac{(\Box C)^2}{\Lambda_C^4 + (\Box C)^2} + \beta_C \frac{X}{\Lambda_X^4 + |X|}, \qquad X = -\tfrac{1}{2}\nabla_\mu C \nabla^\mu C. \tag{6}$$

The first term measures wave-operator activity; the second measures kinetic energy density. Both are bounded in $[0, 1)$ by construction, making $\mathcal{S}_C$ a dimensionless, covariant activity monitor. In the selection functional, $\mathcal{S}_C$ enters with a positive weight λ_{neq}, so configurations with higher dynamical activity are preferred over static or near-vacuum configurations. The activity sector lowers the structural energy for non-trivial but bounded field evolution: $\mathcal{E}$ decreases as $\mathcal{S}_C$ increases, up to the saturation limit imposed by the Λ cutoffs. Throughout this paper, the activity sector is understood to lower the effective structural energy: configurations with larger bounded $\mathcal{S}_C$ are structurally preferred over static ones. The saturation behaviour at large arguments ensures that pathologically active configurations are not infinitely preferred, providing an implicit upper bound on the selected activity level.

3.4 Interaction Connectivity Functional

Without inter-field couplings, no complex physical structure can emerge. For a multi-field theory with fields $\{\Phi_i\}$, a minimal connectivity measure is:

$$\mathcal{N}[\Phi] = \sum_{i<j} \xi_{ij}\, \nabla_\mu \Phi_i\, \nabla^\mu \Phi_j, \tag{7}$$

measuring gradient correlations between field pairs. In the present paper, $\mathcal{N}$ is intentionally kept minimal and local: it measures inter-field coupling structure through gradient correlations only. Larger gradient correlations correspond to larger $\mathcal{N}$, and since the selection functional minimises

$-\mathcal{N}$, configurations with stronger inter-field coupling are preferred. In a single-field baseline, $\mathcal{N} = 0$ by definition.

A more ambitious information-theoretic formulation — for example in terms of mutual-information-type observables over subsystems — is conceptually attractive but mathematically non-trivial in field theory (regulator- and measure-dependent, subsystem-sensitive) and is therefore deferred to future work.

3.5 Consistency Functional

A physical theory must respect its own structural constraints: ghost-freeness, absence of forbidden degrees of freedom, anomaly cancellation, and UV consistency. We encode these as:

$$\Sigma[\Phi] = -\alpha^2 \, \mathcal{A}[\Phi]^2, \tag{8}$$

where $\mathcal{A}[\Phi]$ is a theory-dependent anomaly or constraint-violation measure. $\Sigma = 0$ when all consistency conditions hold exactly. By construction, $\mathcal{B}$ and Σ are orthogonal diagnostics: $\mathcal{B}$ probes flux conservation, Σ probes structural consistency of the effective-theory framework.

Table 1: The five structural diagnostics.

Symbol	Technical name	Physical content	Optimal value
$\mathcal{H}$	Residual Stability Functional	distance from solution manifold	0
$\mathcal{B}$	Conservation Functional	energy–momentum conservation	0
$\mathcal{S}_C$	Dynamical Activity Functional	regulated non-equilibrium activity	high but bounded
$\mathcal{N}$	Connectivity Functional	inter-field coupling structure	maximal
Σ	Consistency Functional	theoretical constraint integrity	0

A remark on the on-shell behaviour of these diagnostics is important. On exact solutions of $\delta S = 0$, the Euler–Lagrange residuals vanish by definition, so $\mathcal{H} = 0$ automatically. Likewise, $\mathcal{B} = 0$ follows from the contracted Bianchi identity on-shell, and $\Sigma = 0$ if all consistency conditions of the theory are satisfied. *On the solution manifold $\mathcal{M}_S$, therefore, the selection principle is primarily driven by $\mathcal{N}$ and $\mathcal{S}_C$*, which measure connectivity and dynamical activity and do not vanish automatically. The diagnostics $\mathcal{H}$, $\mathcal{B}$, and Σ become non-trivially discriminating for approximate, truncated, coarse-grained, or numerically reconstructed solutions — precisely the setting in which the selection functional is most useful as a practical tool. This on-shell/off-shell distinction is a feature of the framework, not a weakness: it reflects the different roles played by exact and approximate solutions in physical modelling.

4 Maximum-Entropy Derivation of the Selection Functional

4.1 Probability Distribution on the Solution Manifold

Rather than postulating the weights of $\mathcal{E}$ by hand, we derive them from a Maximum-Entropy (MaxEnt) principle on the solution manifold $\mathcal{M}_S$.

Define a probability measure $P[\Phi]$ on $\mathcal{M}_S$ with Shannon–Jaynes entropy:

$$\mathcal{S}[P] = -\int_{\mathcal{M}_S} \mathcal{D}\Phi \; P[\Phi] \ln P[\Phi]. \tag{9}$$

We impose three classes of structural constraints, encoding the mean values of the stability,

information, and non-equilibrium sectors:

$$\langle \mathcal{E}_{\text{stab}} \rangle = \bar{E}_{\text{stab}}, \qquad \mathcal{E}_{\text{stab}} = (-\mathcal{H}) + (-\Sigma), \tag{10}$$
$$\langle \mathcal{E}_{\text{cons}} \rangle = \bar{E}_{\text{cons}}, \qquad \mathcal{E}_{\text{cons}} = (-\mathcal{B}), \tag{11}$$
$$\langle \mathcal{E}_{\text{conn}} \rangle = \bar{E}_{\text{conn}}, \qquad \mathcal{E}_{\text{conn}} = -\mathcal{N}, \tag{12}$$
$$\langle \mathcal{E}_{\text{neq}} \rangle = \bar{E}_{\text{neq}}, \qquad \mathcal{E}_{\text{neq}} = (-\mathcal{S}_C). \tag{13}$$

4.2 MaxEnt Solution

Maximising $\mathcal{S}[P]$ subject to normalisation and equations (10)–(13) yields a Boltzmann-type distribution:

$$P[\Phi] = \frac{1}{Z} \exp(-\lambda_{\text{stab}}\, \mathcal{E}_{\text{stab}}[\Phi] - \lambda_{\text{cons}}\, \mathcal{E}_{\text{cons}}[\Phi] - \lambda_{\text{conn}}\, \mathcal{E}_{\text{conn}}[\Phi] - \lambda_{\text{neq}}\, \mathcal{E}_{\text{neq}}[\Phi])\,, \tag{14}$$

with partition function

$$Z = \int_{\mathcal{M}_S} \mathcal{D}\Phi \;\exp(-\lambda_{\text{stab}}\, \mathcal{E}_{\text{stab}} - \lambda_{\text{cons}}\, \mathcal{E}_{\text{cons}} - \lambda_{\text{conn}}\, \mathcal{E}_{\text{conn}} - \lambda_{\text{neq}}\, \mathcal{E}_{\text{neq}})\,. \tag{15}$$

The Lagrange multipliers $\lambda_{\text{stab}}, \lambda_{\text{cons}}, \lambda_{\text{conn}}, \lambda_{\text{neq}} > 0$ are not free parameters but are determined self-consistently by the constraint equations (10)–(13).

4.3 Saddle-Point Approximation and the Selection Principle

In the limit of large λ (strong structural constraints), the distribution (14) concentrates around the saddle point, recovering the variational statement:

$$\Phi_{\text{phys}} = \underset{\Phi \in \mathcal{M}_S}{\arg\min}\, [\lambda_{\text{stab}}\, \mathcal{E}_{\text{stab}}[\Phi] + \lambda_{\text{cons}}\, \mathcal{E}_{\text{cons}}[\Phi] + \lambda_{\text{conn}}\, \mathcal{E}_{\text{conn}}[\Phi] + \lambda_{\text{neq}}\, \mathcal{E}_{\text{neq}}[\Phi]]\,. \tag{16}$$

The minimisation statement in equation (16) is not an independent postulate. It follows as the saddle-point approximation of the MaxEnt distribution (14) in the regime where the structural constraints are sharp, i.e. for sufficiently large positive multipliers λ_i. At finite λ_i, the framework is fundamentally probabilistic rather than purely variational: the selection principle assigns a *distribution* over physical configurations, with the minimiser as the dominant mode.

The master formula (16) admits the compact form:

$$\boxed{\Phi_{\text{phys}} = \underset{\delta S[\Phi]=0}{\arg\min} \Big[\lambda_{\text{stab}} \big(\|R[\Phi]\|^2 + \alpha^2 \mathcal{A}[\Phi]^2\big) + \lambda_{\text{cons}}\, \|\nabla_\mu T^{\mu\nu}\|^2 + \lambda_{\text{conn}}\, (-\mathcal{N}[\Phi]) - \lambda_{\text{neq}}\, \mathcal{S}_C[\Phi]\Big].} \tag{17}$$

As a heuristic summary, one may say that physical reality appears as stabilised information in dynamical geometry. This phrase is intended as an interpretive slogan rather than a formal theorem.

5 Numerical Proof of Concept

To test whether the selection principle actually favours structured, stable configurations, we implement a minimal toy model: a real scalar field $\phi(x,t)$ on a one-dimensional lattice of $N = 20$ sites. The numerical experiment below is not intended as a derivation of a realistic field-theoretic result, but as a minimal proof of concept showing that the selection principle can discriminate between qualitatively different configurations in a reduced setting.

5.1 Dynamical Model

The field satisfies a discretised Klein–Gordon equation with ϕ^4 potential:

$$\ddot{\phi}_i = \frac{\phi_{i+1} + \phi_{i-1} - 2\phi_i}{\Delta x^2} - V'(\phi_i), \qquad V(\phi) = \tfrac{1}{4}(\phi^2 - 1)^2. \tag{18}$$

The dynamics is integrated with a fourth-order Runge–Kutta scheme from random initial conditions.

5.2 Lattice Diagnostics

For each configuration we evaluate four diagnostics.

Residual stability:

$$H_{\text{lat}} = -\frac{\sum_i R_i^2}{\Lambda_H^2 + \sum_i R_i^2}, \qquad R_i = \Delta\phi_i - V'(\phi_i), \tag{19}$$

where $\Delta\phi_i = (\phi_{i+1} + \phi_{i-1} - 2\phi_i)/\Delta x^2$ is the discrete Laplacian.

Connectivity:

$$N_{\text{lat}} = I(\phi_i : \phi_{i+1}), \tag{20}$$

the mutual information between adjacent field values, estimated from their empirical joint distribution. We stress that this is a lattice-specific proxy for inter-site coupling and should not be identified with the continuum gradient-correlation functional $\mathcal{N}$ of Section 3; it serves here only as a numerically convenient connectivity measure for the toy model.

Dynamical activity:

$$S_{\text{lat}} = \frac{\sum_i \pi_i^2}{\Lambda_S^2 + \sum_i \pi_i^2}, \qquad \pi_i = \dot{\phi}_i. \tag{21}$$

Boundary consistency:

$$\Sigma_{\text{lat}} = -\frac{\sum_i \max(0, |\phi_i| - \phi_{\max})^2}{\Lambda_\Sigma^2 + \sum_i \max(0, |\phi_i| - \phi_{\max})^2}, \tag{22}$$

penalising field values outside the admissible range.

5.3 Structural Energy and Selection

The structural energy is

$$E_{\text{lat}} = \lambda_{\text{stab}}[(-H_{\text{lat}}) + (-\Sigma_{\text{lat}})] + \lambda_{\text{conn}}(-N_{\text{lat}}) - \lambda_{\text{dyn}}\, S_{\text{lat}}, \tag{23}$$

and the selected configuration is $\Phi_{\text{phys}} = \arg\min E_{\text{lat}}$.

5.4 Results

Simulations from random initial conditions on the $N = 20$ lattice are evolved for a fixed time using the conservative dynamics (18). We then evaluate E_{lat} for each configuration and identify the energetically preferred state across the ensemble. Since the Klein–Gordon dynamics is conservative and non-dissipative, the field does not dynamically "converge" to an attractor; rather, the selection principle identifies the minimum-E_{lat} configuration post-hoc from the ensemble of dynamically evolved states. The energetically preferred configuration is a *kink* solution $\phi(x) \approx \tanh(x/\xi)$, which interpolates between the two minima of V. The kink maximises the inter-site gradient correlation $\mathcal{N}$ while exhibiting non-trivial but bounded dynamical activity,

whereas the homogeneous vacua $\phi = \pm 1$ score significantly lower on connectivity. Typical diagnostic values for this preferred configuration are:

$$H_{\text{lat}} \approx -0.07, \qquad N_{\text{lat}} \approx 1.2 \text{ bits},$$
$$S_{\text{lat}} \approx 0.5, \qquad \Sigma_{\text{lat}} \approx 0. \tag{24}$$

These values indicate: high solution-manifold proximity (small residual), high connectivity between adjacent sites, moderate dynamical activity, and no boundary violations. The homogeneous configurations $\phi = \pm 1$ (vacuum states) score lower on connectivity and dynamical activity, and are disfavoured by the selection principle.

This result should be interpreted only as a proof of concept for the discriminative capacity of the selection functional, not as evidence for universal topological selection. The toy model is too reduced to support stronger conclusions.

This result is consistent with the hypothesis that the structural selection principle favours ordered, connected, moderately active configurations over trivial or chaotically unstable ones. We emphasise that this toy model is not a realistic cosmology; it is a minimal proof of concept for the selection mechanism.

6 FLRW Reduction of the Structural Diagnostics

As a first step toward connecting the general framework to cosmological physics, we reduce the five structural diagnostics to a flat Friedmann–Lemaître–Robertson–Walker (FLRW) background. This is not intended as a derivation of new cosmological results, but as a background-level consistency analysis illustrating how the selection functional acquires concrete physical content in a familiar setting.

6.1 Setup

We consider the spatially flat FLRW metric

$$ds^2 = -dt^2 + a(t)^2\, \delta_{ij}\, dx^i dx^j, \tag{25}$$

with a single scalar field $C = C(t)$ (homogeneous background), coupled non-minimally to gravity via $F(C) = 1 + \xi C/M_C$. The matter sector contributes an energy density ρ_m and pressure p_m. The background Friedmann and Klein–Gordon equations are:

$$3M_{\text{Pl}}^2 F(C)\, H^2 = \rho_m + \tfrac{1}{2}\dot{C}^2 + V(C) - 3M_{\text{Pl}}^2 H\dot{F}, \tag{26}$$
$$\ddot{C} + 3H\dot{C} + V'(C) = \tfrac{1}{2}M_{\text{Pl}}^2 F'(C)\, R, \tag{27}$$
$$\dot{\rho}_m + 3H(\rho_m + p_m) = 0. \tag{28}$$

6.2 FLRW Reduction of the Five Diagnostics

We now evaluate the five diagnostics on the FLRW background.

Residual Stability ($\mathcal{H}$). Defining the Friedmann and Klein–Gordon residuals

$$R_F \equiv 3M_{\text{Pl}}^2 F H^2 - \rho_m - \tfrac{1}{2}\dot{C}^2 - V(C) + 3M_{\text{Pl}}^2 H\dot{F}, \tag{29}$$
$$R_{KG} \equiv \ddot{C} + 3H\dot{C} + V'(C) - \tfrac{1}{2}M_{\text{Pl}}^2 F'(C)\, R, \tag{30}$$

the FLRW residual stability functional becomes

$$\mathcal{H}_{\text{FLRW}} = -(R_F^2 + R_{KG}^2). \tag{31}$$

$\mathcal{H}_{\text{FLRW}} = 0$ if and only if both background equations are exactly satisfied.

Conservation Functional ($\mathcal{B}$). The matter energy–momentum non-conservation reduces to

$$\mathcal{B}_{\text{FLRW}} = -\left(\dot{\rho}_m + 3H(\rho_m + p_m)\right)^2 . \tag{32}$$

On exact solutions, $\mathcal{B}_{\text{FLRW}} = 0$ by the contracted Bianchi identity. As with the covariant $\mathcal{B}$, this diagnostic is most informative for approximate or truncated backgrounds.

Dynamical Activity ($\mathcal{S}_C$). In FLRW with a homogeneous field, $\Box C = -\ddot{C} - 3H\dot{C}$ and $X = \frac{1}{2}\dot{C}^2$. The full activity functional reduces to a proxy depending on $\dot{C}$:

$$\mathcal{S}_{\text{FLRW}} \approx \eta_C \frac{(\ddot{C} + 3H\dot{C})^2}{\Lambda_C^4 + (\ddot{C} + 3H\dot{C})^2} + \beta_C \frac{\frac{1}{2}\dot{C}^2}{\Lambda_X^4 + \frac{1}{2}\dot{C}^2}. \tag{33}$$

This is a simplified background-level proxy for the full covariant activity functional; the two expressions agree when evaluated on homogeneous FLRW configurations. Static configurations ($\dot{C} = 0$) yield $\mathcal{S}_{\text{FLRW}} = 0$ and are therefore not favoured by the activity sector of the selection functional.

Connectivity ($\mathcal{N}$). In the single-field FLRW reduction,

$$\mathcal{N}_{\text{FLRW}} = 0, \tag{34}$$

as expected for a single homogeneous background field. $\mathcal{N}$ becomes non-trivial only in multi-field or inhomogeneous extensions.

Consistency Functional (Σ). Ghost-freeness requires $F(C) > 0$. Rather than measuring the distance from $F = 1$, which would conflate deviation from GR with inconsistency, we formulate Σ as a soft barrier penalising configurations near $F(C) \to 0^+$:

$$\Sigma_{\text{FLRW}} = -\frac{\alpha^2}{F(C)^2 + \epsilon}, \tag{35}$$

where $\epsilon > 0$ is a small regulator. $\Sigma_{\text{FLRW}} \to -\infty$ as $F(C) \to 0$, providing a strong penalty for ghost-near configurations, while $\Sigma_{\text{FLRW}} \approx -\alpha^2$ for $F(C) \sim \mathcal{O}(1)$, contributing a nearly constant offset. This formulation correctly identifies ghost-freeness, not proximity to GR, as the relevant consistency condition.

6.3 Qualitative Selection Tendencies

Collecting the FLRW reductions, the structural energy on a homogeneous scalar–tensor background reads:

$$\mathcal{E}_{\text{FLRW}} = \lambda_{\text{stab}}(-\mathcal{H}_{\text{FLRW}} - \Sigma_{\text{FLRW}}) + \lambda_{\text{cons}}(-\mathcal{B}_{\text{FLRW}}) + \lambda_{\text{conn}}(-\mathcal{N}_{\text{FLRW}}) - \lambda_{\text{neq}}\,\mathcal{S}_{\text{FLRW}}. \tag{36}$$

Minimising $\mathcal{E}_{\text{FLRW}}$ over backgrounds suggests the following *qualitative selection tendencies* (not rigorous results):

1. **Exact solutions preferred.** Configurations satisfying both Friedmann and Klein–Gordon equations exactly ($R_F = R_{KG} = 0$) are strongly preferred over approximate ones.

2. **Singular and ghost-near backgrounds disfavoured.** As $F(C) \to 0^+$ or the field reaches a ghost configuration, $\Sigma_{\text{FLRW}} \to -\infty$ strongly penalises the background.

3. **Dynamically active backgrounds preferred.** Static or near-vacuum configurations ($\dot{C} \approx 0$) yield $\mathcal{S}_{\text{FLRW}} \approx 0$ and are disfavoured relative to backgrounds with non-trivial field evolution.

4. **Pure de Sitter vs. rolling-field backgrounds.** A pure de Sitter solution with $\dot{C} = 0$ and $H = \text{const}$ scores zero on dynamical activity, whereas a rolling-field background with $\dot{C} \neq 0$ and moderate kinetic energy is preferred by $\mathcal{S}_{\text{FLRW}}$.

We stress that these are qualitative tendencies derived from the structure of the functional, not quantitative predictions. A rigorous minimisation over the space of FLRW backgrounds would require specifying the Lagrange multiplier values λ_i and the potential $V(C)$, which are left as open parameters at this stage.

6.4 Three FLRW Benchmark Cases

To illustrate the discriminative power of the selection functional concretely, we evaluate $\mathcal{E}_{\text{FLRW}}$ for three representative cosmological backgrounds. All cases use the benchmark parameters $\xi = 10^{-3}$, $M_C = M_{\text{Pl}}$, $H_0 = 10^{-3} M_{\text{Pl}}$, $\lambda_i = 1$, and identical matter equation of state $w_m = -1$. Equal weights $\lambda_i = 1$ are chosen as an illustrative benchmark; in future work, the λ_i will be fixed through information-balance conditions or cosmological complexity measures rather than set by hand. Cases A and B differ only in the scalar-field dynamics; Case C is a deliberately constructed ghost-near stress test and should not be interpreted as a realistic cosmological background. The numerical values are computed from the Python implementation described in the companion code [1].

Case A: Exact de Sitter. $H = H_0 = \text{const}$, $\dot{C} = 0$, with $C = 6\xi M_C \approx 0.006\, M_C$ chosen as a benchmark-calibrated static value that approximately satisfies the background scalar equation for the selected parameter set. Both background equations are therefore approximately satisfied. This yields $\mathcal{H}_A \approx 0$, $\mathcal{B}_A \approx 0$, but $\mathcal{S}_A = 0$: no dynamical activity.

Case B: Slow-roll scalar–tensor FLRW. $H \approx H_0$, $\dot{C} \neq 0$ with $|\xi C/M_C| \ll 1$, same matter content as Case A. The field evolves with a slight deviation from exact slow-roll ($\ddot{C} \neq -3H\dot{C}$), ensuring both activity terms in $\mathcal{S}_C$ contribute. Background equations are satisfied with small residuals. This yields $\mathcal{H}_B \approx 0$, $\mathcal{B}_B \approx 0$, and $\mathcal{S}_B > 0$: non-trivial but bounded dynamical activity.

Case C: Ghost-near stress test. $F(C) \approx 0.05 \ll 1$, deliberately constructed to lie near the ghost-crossing boundary. This is not a realistic cosmological background; it is included to verify that the Σ term correctly penalises structurally inconsistent configurations.[1]

The resulting benchmark values are summarised in Table 2.

Table 2: Structural energy benchmark for three FLRW backgrounds ($\xi = 10^{-3}$, $w_m = -1$, $\lambda_i = 1$). Lower $\mathcal{E}$ = structurally preferred. Case A uses the KG-compatible static value $C = 6\xi M_C$. Case C is a ghost-near stress test, not a realistic background.

Case	$F(C)$	$\mathcal{H}$	$\mathcal{B}$	$\mathcal{S}_C$	Σ	$\mathcal{E}$
A: de Sitter	1.000	0.000	0.000	0.000	−1.000	1.000
B: slow-roll	1.0001	0.000	0.000	0.111	−1.000	0.889
C: ghost-near*	0.050	−0.204	0.000	0.111	−399.840	399.933

* Deliberately constructed stress test; not a realistic cosmological solution.

The ranking $\mathcal{E}_B < \mathcal{E}_A < \mathcal{E}_C$ is consistent with the structural selection hypothesis. Cases A and B have identical matter content; their difference arises solely from the scalar field activity $\mathcal{S}_C$, which favours the dynamically active slow-roll background. Case C demonstrates that the Σ barrier correctly identifies and penalises ghost-near configurations. We stress that this benchmark tests the *discriminative capacity* of the functional, not its ability to derive cosmological predictions from first principles. The purpose of the FLRW benchmark is not to derive cosmology from first principles, but to demonstrate that the structural functional is capable of ranking controlled background classes in a reproducible way.

[1] A physically motivated near-ghost configuration arises naturally in the CEST scalar–tensor model when $\xi C/M_C \to -1$, i.e. when the coherence field drives the non-minimal coupling toward zero. See the companion paper [2] for details.

7 Discussion

7.1 Relation to Existing Frameworks

The two-level structure proposed here is related to, but distinct from, several known approaches.

Relation to scalar–tensor cosmology. A concrete realisation of the diagnostic framework introduced here — applied to a specific scalar–tensor model with coherence-field coupling $F(C) = 1 + \xi C/M_C$ — is developed in a companion paper [2]. That work demonstrates how the five structural diagnostics can be evaluated explicitly on cosmological solutions and connected to observational constraints on G_{eff} variation. The present paper provides the general meta-theoretic foundation; the companion paper provides the concrete field-theoretic realisation.

Variational principles. The dynamics $\delta S = 0$ is itself a selection principle, but one that typically leaves a large degenerate solution space. Our structural selection operates *within* this space and is therefore logically posterior to the variational principle.

Anthropic and landscape approaches. The string landscape and anthropic arguments also address solution multiplicity, but typically invoke observer selection or statistical weighting over a multiverse. Our approach is purely structural and observer-independent.

Stability selection. In dynamical systems theory, Lyapunov stability provides a criterion for which equilibria are physical. Our Residual Stability Functional is the field-theoretic generalisation of this idea.

MaxEnt statistical mechanics. The Maximum-Entropy derivation of equation (14) is formally identical to the derivation of the Boltzmann distribution in statistical mechanics, with the solution manifold playing the role of phase space and the structural sectors playing the role of conserved quantities.

7.2 Open Questions

The framework presented here is a research programme rather than a complete theory. The following questions remain open:

1. **Determination of λ values.** The Lagrange multipliers $\lambda_{\text{stab}}, \lambda_{\text{cons}}, \lambda_{\text{conn}}, \lambda_{\text{neq}}$ must be fixed by physical constraints or further theoretical arguments. Their origin from a deeper principle is not yet understood.

2. **Definition of $\mathcal{N}$ in field theory.** The mutual-information version of the connectivity functional requires a well-defined measure on the space of field configurations, which is non-trivial in QFT.

3. **Physical predictions.** Whether the selection principle produces new, testable predictions beyond those of the underlying dynamical theory is the central open question. Candidate applications include singularity avoidance, information conservation, and complexity emergence.

4. **Uniqueness.** Whether $\mathcal{E}[\Phi]$ has a unique minimiser, or a family of degenerate minima, depends sensitively on the structure of $\mathcal{M}_S$ and the functional form of the diagnostics.

5. **Quantum extension.** In quantum field theory, the solution manifold is replaced by a space of quantum states. The extension of the selection principle to this setting requires separate investigation.

7.3 Philosophical Implications

If the framework proves consistent, it suggests that physical reality is not merely any solution of the fundamental equations, but the *most stable*, *most connected*, and *most active* solution consistent with theoretical integrity. In this sense, the selection principle functions as a *meta-principle of physical existence*:

A system can stably exist only if it is coherent, conserved, dynamically active, connected, and self-consistent.

We stress that this is a structural, not a teleological, statement: the universe does not "strive" toward MAAT structure; rather, wherever stable structure exists, it has this form.

8 Conclusion

We have proposed a two-level framework for physical solution selection in field theory. The fundamental dynamics $\delta S[\Phi] = 0$ defines the space of possible configurations; a structural selection functional $\mathcal{E}[\Phi]$, constructed from five covariant diagnostics and derived from a Maximum-Entropy principle on the solution manifold, identifies the physically preferred configurations.

The five diagnostics — Residual Stability, Conservation Integrity, Dynamical Activity, Connectivity, and Theoretical Consistency — each vanish or attain their optimal value on exact, well-behaved solutions, and collectively measure the distance of a configuration from the space of physically viable field theories.

A numerical proof of concept on a one-dimensional scalar lattice confirms that the selection principle favours topologically structured, connected, moderately active configurations, consistent with the hypothesis.

The framework is presented as a research programme. Its primary contribution is the formulation of a mathematically grounded, observer-independent selection principle that complements, rather than replaces, the dynamical equations. The reproducible FLRW benchmark further shows that the proposed functional can rank controlled cosmological background classes in a way consistent with its intended structural interpretation. Future work will address the determination of the Lagrange multiplier values, the quantum extension, and the derivation of concrete new observational predictions.

A Reproducibility

The numerical benchmark of Section 6.4 is fully reproducible from the companion Python code available at https://github.com/Chris4081/structural-selection-principle.

The benchmark parameters are:

$$\begin{aligned} &\xi = 10^{-3}, \quad M_C = M_{\rm Pl}, \quad H_0 = 10^{-3}\, M_{\rm Pl}, \quad w_m = -1, \\ &\eta_C = \beta_C = 1, \quad \Lambda_C = \Lambda_X = 10^{-2} \quad \text{(benchmark cutoffs in normalised Planck units)}, \\ &\lambda_{\rm stab} = \lambda_{\rm cons} = \lambda_{\rm conn} = \lambda_{\rm neq} = 1, \quad \alpha = 1, \quad \epsilon = 10^{-6}. \end{aligned}$$

Case A uses $C = 6\xi M_C \approx 0.006\, M_C$ (KG-compatible static value); Case B uses $C = 0.1\, M_C$, $\dot{C} = 0.05\, H_0 M_C$, $\ddot{C} = -2.5\, H\dot{C}$; Case C sets $F(C) \approx 0.05$ (ghost-near stress test).

The code requires only `numpy` and runs without additional dependencies:

```
pip install numpy
python structural_selection_flrw_v4.py
```

References

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